

Stat 5550 Final Project

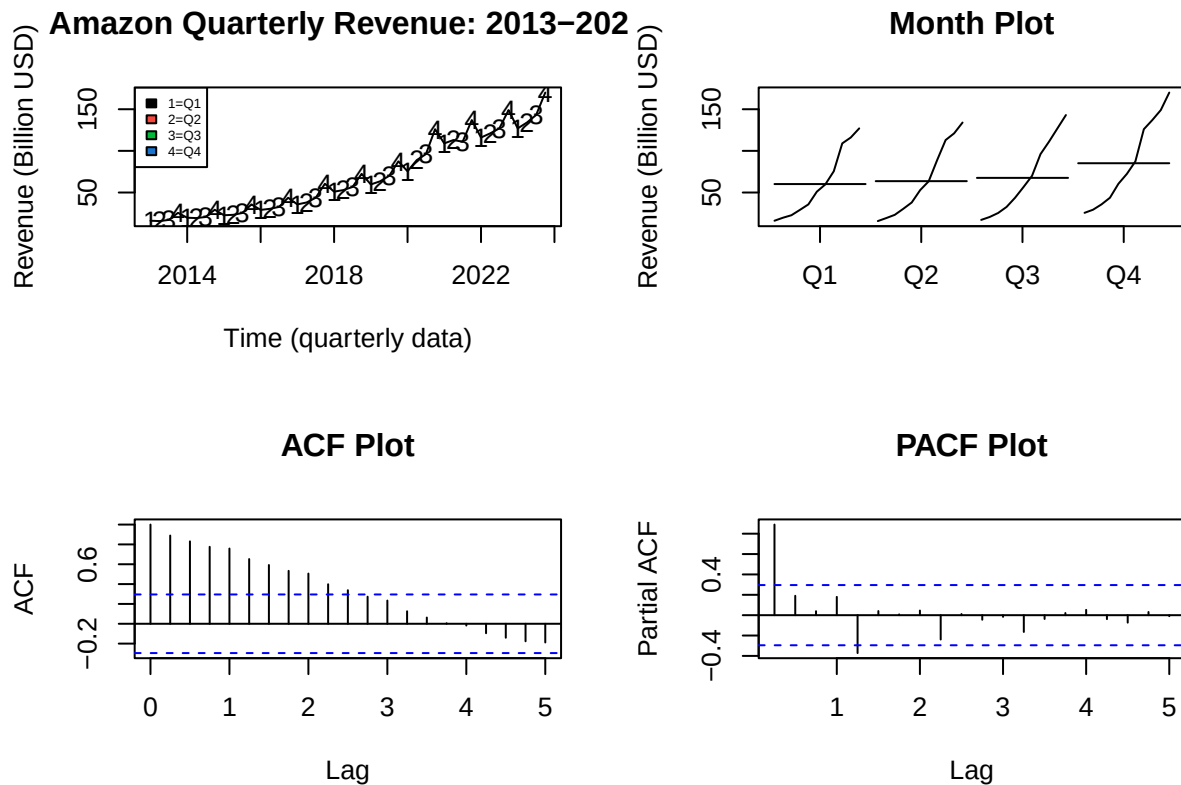
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2025-04-22

Introduction

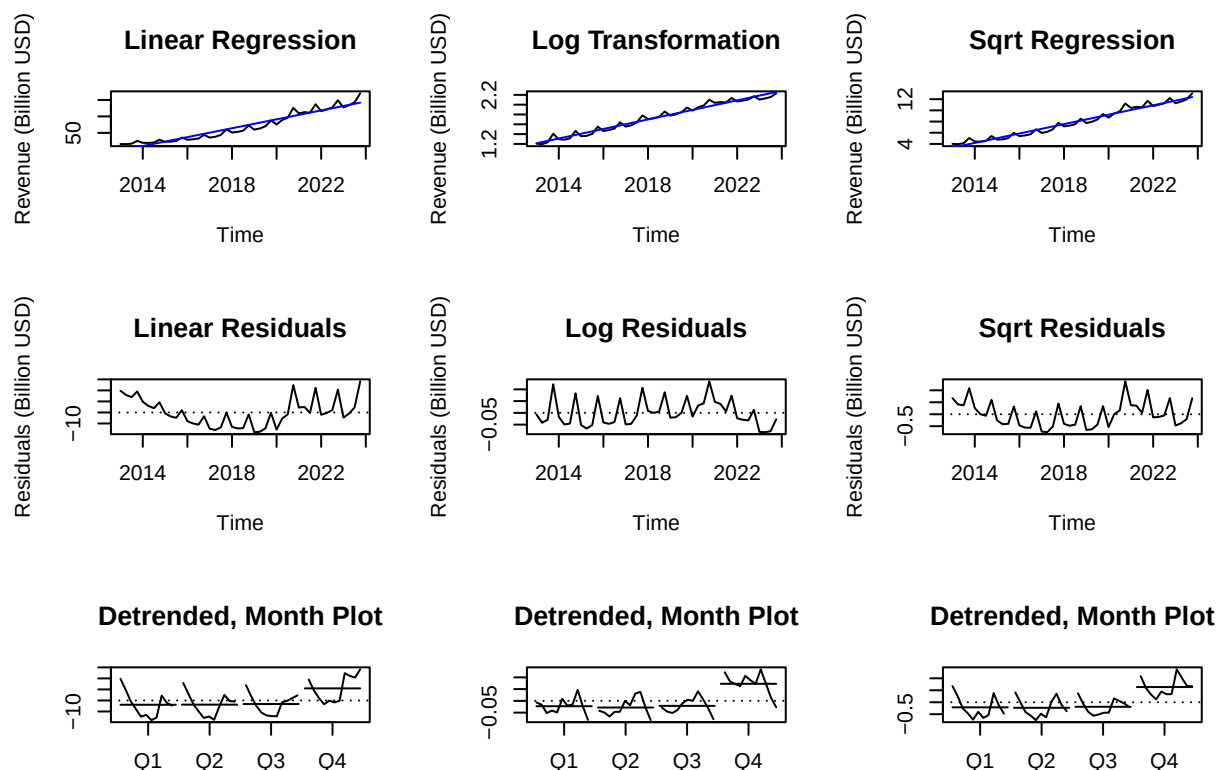
The data set used for this analysis is called “Amazon.csv” from the website <https://www.kaggle.com/datasets/patricklford/amazon-alibaba-and-ebay-2013-2023?select=Amazon.csv>. It contains the quarterly revenue in billions of USD of Amazon from the first quarter of 2013 (2013 Q1) all the way through the fourth/last quarter of 2023 (2023 Q4). This data set has 44 observations over a 11 year period. It capture annual seasonality and shows an increasing long-term growth trend.

The time series x_t displays the raw time series with an upward trend in revenue over the 11 year period, with some sharp increases around 2018-2023 in the last quarter (4). There seems to also be a periodic or seasonal pattern where there is higher revenue typically during the fourth quarter of each year, which is probably due to holidays and other events that are going on around the winter months.



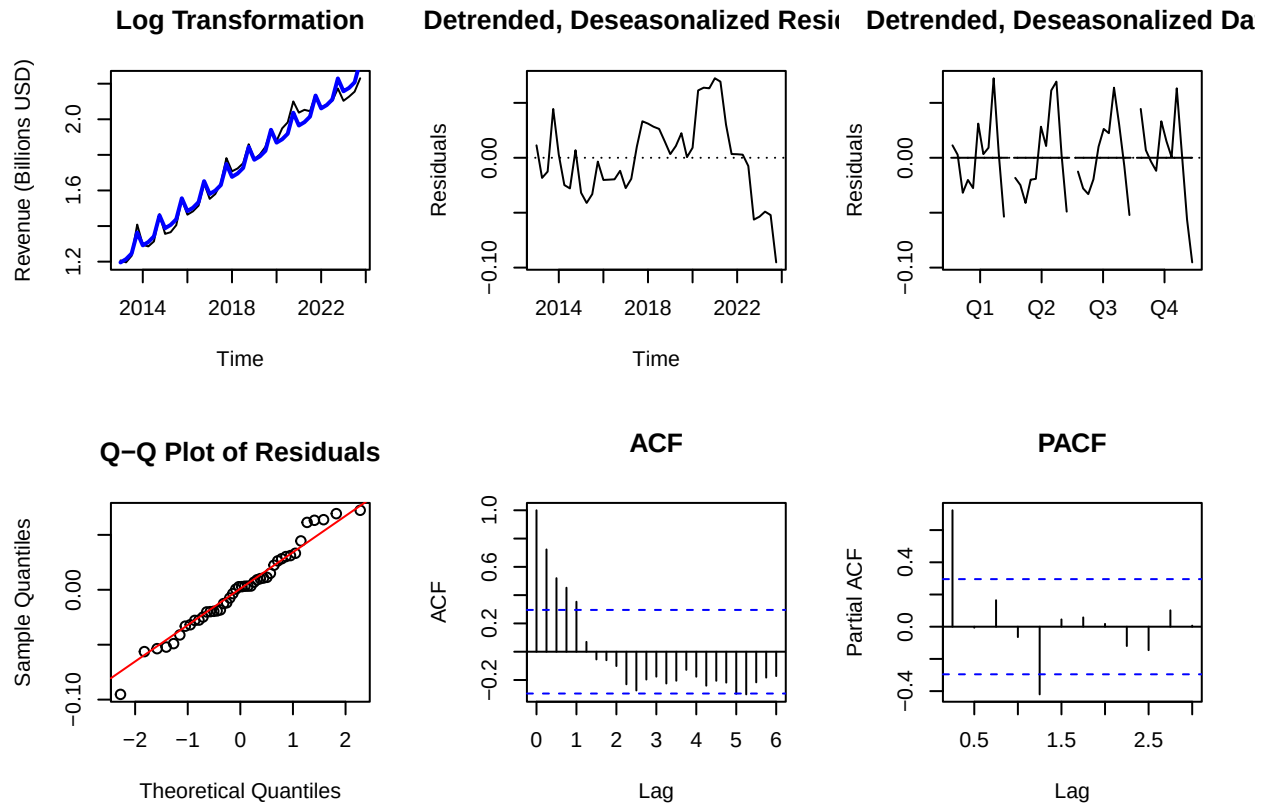
Transformations

The linear-transformation $x_t = \beta_0 + \beta_1 t + w_t$, where $w_t \sim wn(0, \sigma^2)$ were applied, but based on the plots shown, the log-transformation series $y_t - \log x_t$ seems to approximate the data better to de-trend and get rid of any variance, where the de-trended residual seems to be more appropriate, but based on the month plots (quarterly residuals), there seems to still be some seasonality behavior remaining in the data.



Seasonal Means

Below I use the seasonal means to get rid of any seasonality that is the time series shown in the month plot from above.



From the month plot it can be seen that the seasonality has been removed, the qq plots are mostly in the red diagonal line, and the ACF decreases exponentially, I also tested using the Ljung-Box test to make sure that this is stationary but not white noise. Based on the p-value < 0.05 , we can reject the null hypothesis that the plot is white noise.

```
##
## Box-Ljung test
##
## data: amz
## X-squared = 67.116, df = 12, p-value = 1.105e-09
```

Possible ARMA models:

Based on the values above, it seems that the possible $MA(q)$ are at spikes 0, 0.25, 0.50, 0.75, 1, 5, 5.25 (lag=0,1,2,3,4,20,21) and for $AR(p)$ are at spikes 0, 0.25, 1.25 (lag=0,4)

##	Model	arphi_1	phi_2	phi_3	phi_4
## 1	ARMA(0,0)				
## 2	ARMA(0,1)				
## 3	ARMA(0,2)				
## 4	ARMA(0,3)				
## 5	ARMA(0,4)				
## 6	ARMA(4,0)	0.9719(0.1580)	-0.4012(0.2145)	0.3948(0.2175)	-0.1027(0.1705)
## 7	ARMA(4,1)	0.1944(0.1649)	0.3666(0.1562)	-0.2077(0.1514)	0.4846(0.1520)
## 8	ARMA(4,2)	0.1473(0.1626)	0.1887(0.1469)	0.1312(0.1615)	0.5390(0.1541)

```

## 9  ARMA(4,3) -0.2678(0.1974)  0.1064(0.1581)  0.1621(0.1601)  0.6310(0.1671)
## 10 ARMA(4,4) -0.0189(0.1348) -0.2379(0.1207)  0.0413(0.1344)  0.6653(0.1305)
##      matheta_1      theta_2      theta_3      theta_4      AIC
## 1
## 2  0.6636 (0.0855)
## 3  1.3074 (0.1132) 0.8193 (0.0900)
## 4  1.2957 (0.1445) 0.7956 (0.1992) -0.0176 (0.1331)
## 5  0.9642 (0.1374) 0.5763 (0.1785)  0.482 (0.190) 0.7077 (0.1362) -4.843923
## 6
## 7  0.7982(0.1147)
## 8  1.3729(0.1548)  0.7487(0.1330)
## 9  1.4917(0.2338)  1.0045(0.3014)  0.2188(0.2009)
## 10 1.1096(0.1343)  1.2172(0.1588)  1.1636(0.1828)  0.8735(0.1489) -5.060983

```

AIC for ARMA where q=20,21:

```

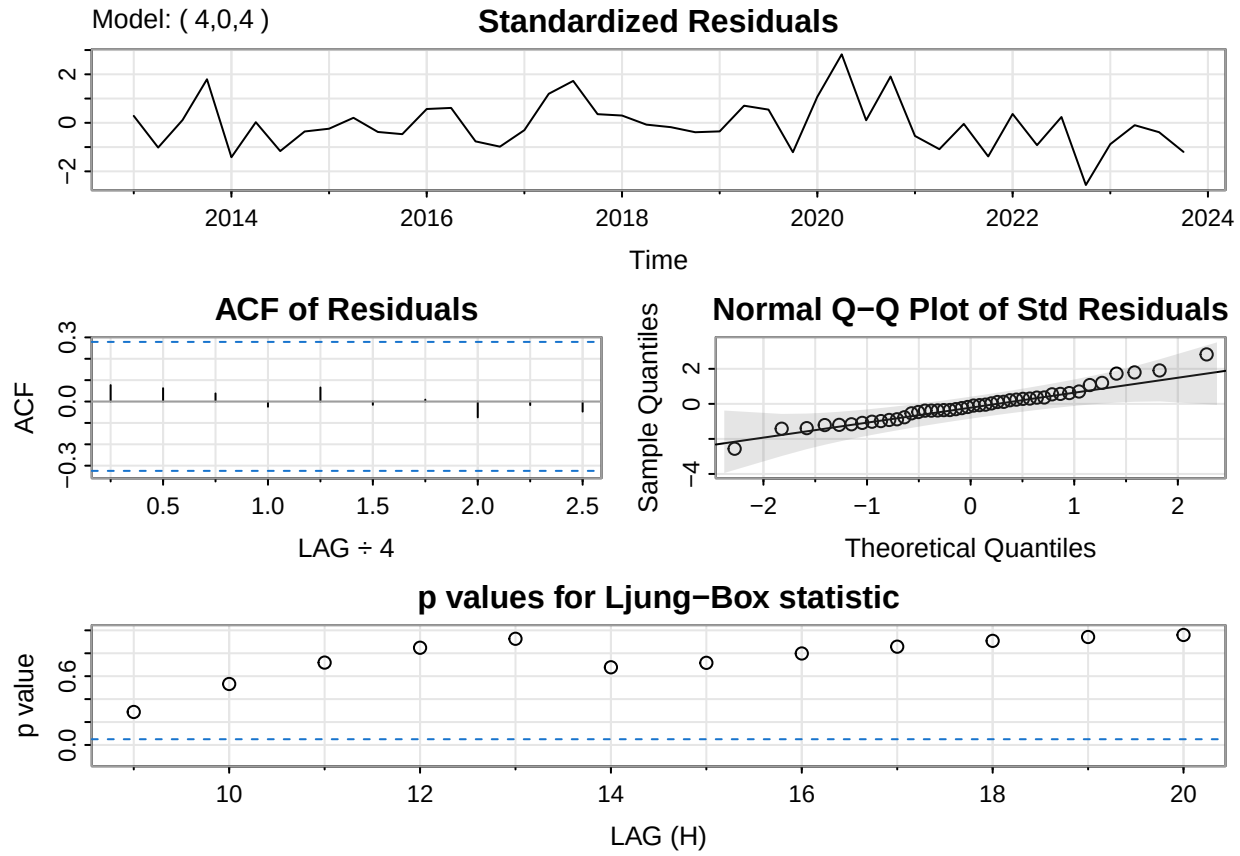
##      Model      AIC
## 1 ARMA(0,20) -4.729919
## 2 ARMA(0,21) -4.685529
## 3 ARMA(4,20) -4.685529
## 4 ARMA(4,21) -4.685529

```

From these models, the lowest AIC is -5.060983 for ARMA(4,4), thus the estimated model would be:

$$z_t = -0.0189z_{t-1} - 0.2379z_{t-2} + 0.0413z_{t-3} + 0.6653z_{t-4} + 1.1096w_{t-1} + 1.2172w_{t-2} + 1.1636w_{t-3} + 0.8735w_{t-4} + w_t$$

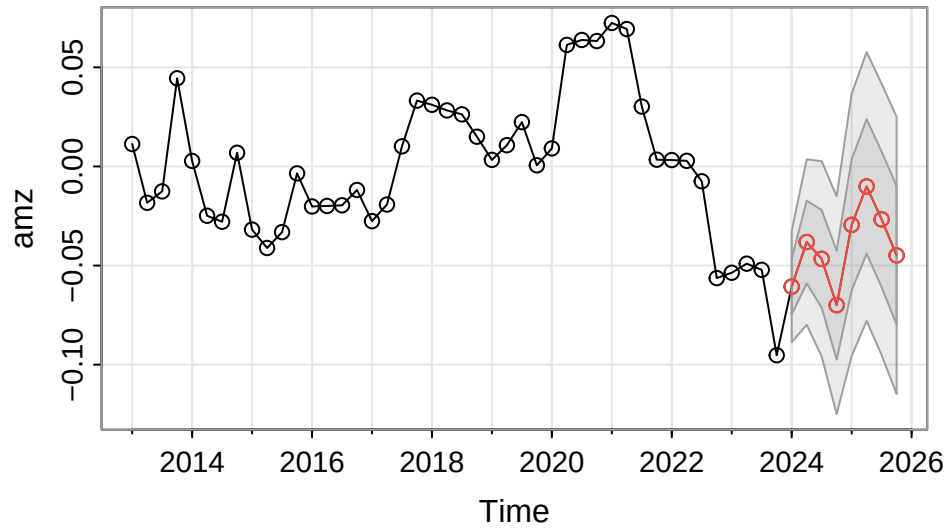
, where $w_t \stackrel{\text{iid}}{\sim} N(0, 0.000188888)$. Below is the standardized residuals. The model seems to fit well: the ACF displays no large auto-correlations, the residuals appears to be normally distributed, and the Ljung-Box test indicates no evidence against iid noise.



Commentary:

- ARMA(0,0): The model is too simplistic and does not capture all ARMA models, and has the highest AIC values, the diagnostics shows that there is still ACF and residual patterns.
- ARMA(0,1): The MA1 (0.6636) is significant and its standard error (0.0855) is robust.
- ARMA(0,2): This model is more significant but it still has a high AIC. The first 3 p-values are not above the interval.
- ARMA(0,3): This model is not significant, the estimate is in the confidence interval (0.1331) and AIC (-4.710342) is worst than ARMA(0,2).
- ARMA(0,4): This model seems significant but the plots show overfitting.
- ARMA(0,20): This model has a good fit, but there might be overfitting.
- ARMA(0,21): This model's AIC (-4.685529) is worst than ARMA(0,20).
- ARMA(4,0): The AR's show varying significance. The AIC (-4.607089) is also really high. The residual autocorrelation remains.
- ARMA(4,1): This model's residuals suggest some overfitting.
- ARMA(4,2): The MA(1) and MA(2) are significant, and the plot shows good residual behavior.
- ARMA(4,3): The MA are significant, but the standard errors not much and the AIC is worst than ARMA(4,2).
- ARMA(4,20): p-values below interval
- ARMA(4,21): p-values below interval

ARMA Forecast



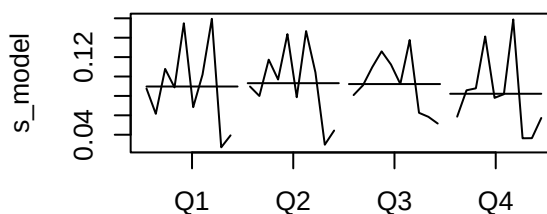
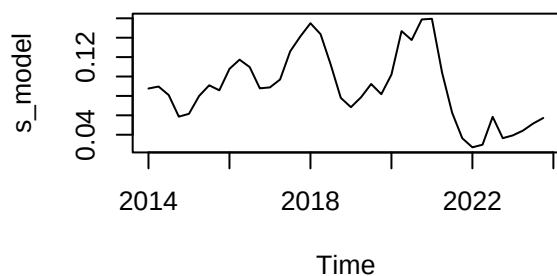
##	Year_Quarter	Forecast	Lower_95_CI	Upper_95_CI
## 1	2024 Q1	-0.06061778	-0.08815033	-0.033085219
## 2	2024 Q2	-0.03812561	-0.07904349	0.002792263
## 3	2024 Q3	-0.04662377	-0.09488391	0.001636366
## 4	2024 Q4	-0.06998172	-0.12381481	-0.016148633
## 5	2025 Q1	-0.02948845	-0.09427294	0.035296040
## 6	2025 Q2	-0.01008362	-0.07654068	0.056373432
## 7	2025 Q3	-0.02670316	-0.09378764	0.040381323
## 8	2025 Q4	-0.04487160	-0.11332846	0.023585256

The plot shows the forecast of 8 quarters (2 years) of revenue using ARMA(4,4) model, and the table shows the prediction intervals for each of the forecasted quarters.

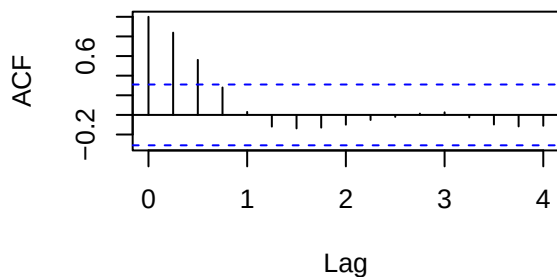
SARIMA Modeling

Taking the log-transformation of the time series again, the SARIMA model will be applied. To de-seasonalize, differencing will be used.

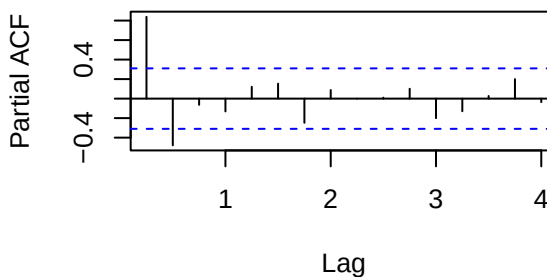
Detrended, Differenced Data



Series s_model



Series s_model



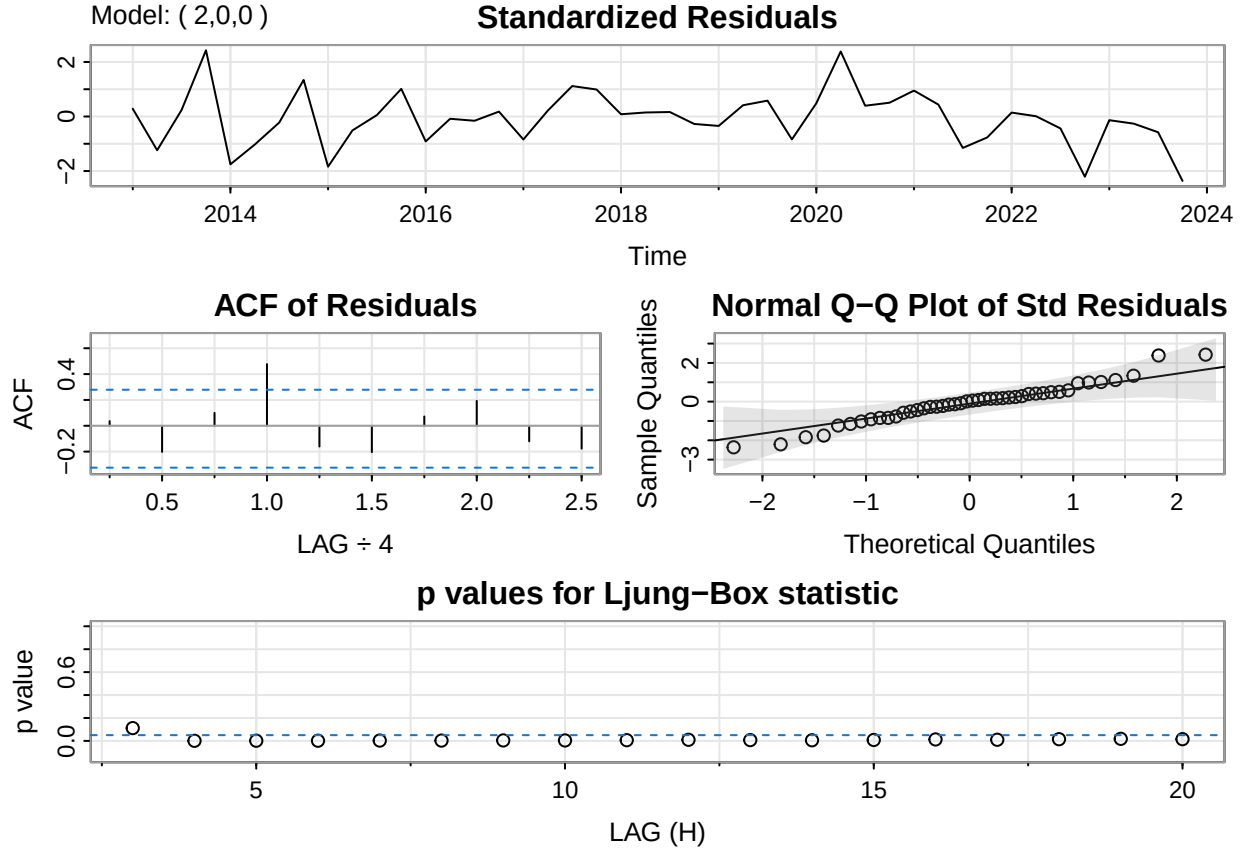
Based on the values above, it seems that the possible $MA(q)$ are at spikes 0, 0.25, 0.50 ($lag=0,1,2$) and for $AR(p)$ are at spikes 0, 0.25, 0.50 ($lag=0,1,2$). The Q values is only at 0, and P value is at 0 only as well. This model is differenced once (quarter-4), so D is 1, with s , periodicity of 4.

##	Model	arphi_1	phi_2	matheta_1	theta_2
## 1	ARMA(0,0)				
## 2	ARMA(0,1)			1e+00(0.0754)	
## 3	ARMA(0,2)			1.3861(0.1720)	0.3861(0.1328)
## 4	ARMA(1,0)	0.7597(0.1010)			
## 5	ARMA(1,1)	0.5303(0.1374)		0.9999(0.1212)	
## 6	ARMA(1,2)	0.5444(0.1896)		0.9798(0.2261)	-0.0202(0.1927)
## 7	ARMA(2,0)	1.2666(0.1228)	-0.6342(0.1202)		
## 8	ARMA(2,1)	0.5142(0.1675)	0.0279(0.1662)	1.0000(0.1142)	
## 9	ARMA(2,2)	-0.0627(0.4791)	0.3969(0.2655)	1.5294(0.4827)	0.5295(0.4767)
##	AIC				
## 1	-2.98464				
## 2	-3.954026				
## 3	-4.117597				
## 4	-3.823539				
## 5	-4.22994				
## 6	-4.174687				
## 7	-4.306731				
## 8	-4.175167				
## 9	-4.138213				

From these models, the lowest AIC is -4.306731 for ARMA(2,0), thus the estimated model would be:

$$(1 - \phi_1 B)(1 - \phi_2 B^4) \nabla_4^1 \nabla_1^1 x_t = w_t$$

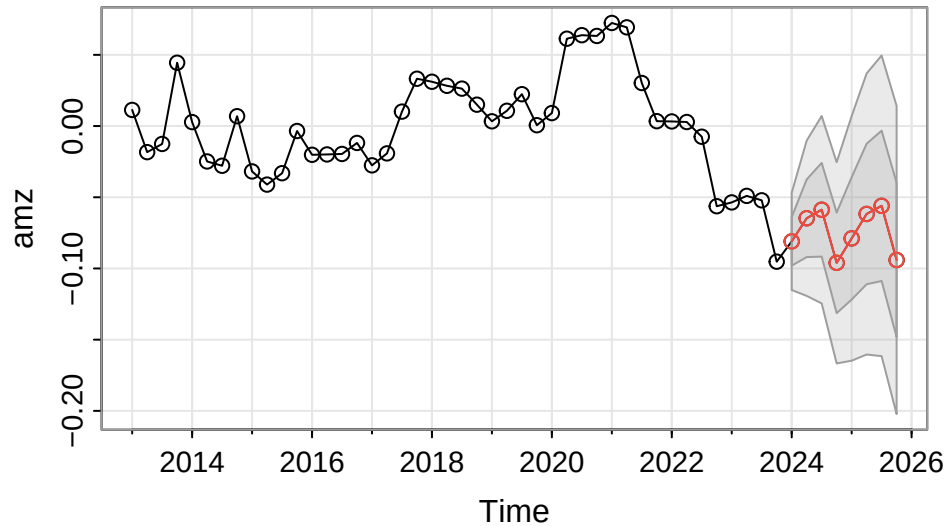
, where $w_t \stackrel{\text{iid}}{\sim} N(0, 0.0005986009)$. Below is the standardized residuals. The model seems to fit well: the ACF displays no large auto-correlations, the residuals appears to be normally distributed, and the Ljung-Box test indicates no evidence against iid noise.



Commentary:

- All models had p-values below the interval line, so the lowest AIC was for ARMA(2,0).
- ARMA(0,0): AIC is the worst.
- ARMA(0,1): The MA is significant, but p-value is below interval.
- ARMA(0,2): Same as ARMA(0,1)
- ARMA(1,0): AR's are significant.
- ARMA(1,1): Same as ARMA(1,0)
- ARMA(1,2): Same as ARMA(1,0)
- ARMA(2,1): Same as ARMA(0,1)
- ARMA(2,2): MA (0.5295) is not significant because of the high standard error (0.4767).

SARIMA Forecast



##	Year_Quarter	Forecast	Lower_95_CI	Upper_95_CI
## 1	2024 Q1	-0.08099181	-0.1144590	-0.047524624
## 2	2024 Q2	-0.06473638	-0.1181812	-0.011291555
## 3	2024 Q3	-0.05876843	-0.1232483	0.005711402
## 4	2024 Q4	-0.09605875	-0.1653001	-0.026817441
## 5	2025 Q1	-0.07889066	-0.1630548	0.005273490
## 6	2025 Q2	-0.06173211	-0.1584506	0.034986394
## 7	2025 Q3	-0.05602031	-0.1594796	0.047438953
## 8	2025 Q4	-0.09405612	-0.2000909	0.011978628

The plot shows the forecast of 8 quarters (2 years) of revenue using SARIMA(2,0,0)x(0,1,0) model, and the table shows the prediction intervals for each of the forecasted quarters.

Model Comparison

Compare the models you fit in parts (2) and (3) above. How are they similar? How are they different? Which model do you prefer, and why?

Both of the ARMA and SARIMA models try to capture the time series autocorrelation structure. They both need to be de-trended and de-seasonalized to be able to be modeled.

Some of the differences of these two models is that the ARMA models have to manually de-seasonalize the time series, while the SARIMA model incorporates seasonal differencing (D), which explicitly accounts for periodic patters. For ARMA models, they also rely only on AR and MA terms, which might not capture the seasonality as effectively. The ARMA model is more complex, which might lead to overfitting. It can also be seen that the AIC of the two values are -5.060983 (ARMA) and -4.306731 (SARIMA), even though ARMA AIC is lower, SARIMA models is preferred if the series exhibits significant seasonality that ARMA models overlook.

I think I would prefer the SARIMA(2,0) model if the series show clear seasonality patters, since this model explicitly accounts for seasonality in its structure.

Conclusion

The analysis of Amazon's quarterly revenue data from 2013 Q1 to 2023 Q4 revealed distinct trends and seasonal patterns. The time series exhibited long-term growth in Amazon's revenue, with noticeable spikes in the fourth quarter of each year. The beginning exploration showed strong evidence of seasonality and upward trend, so it was necessary to make transformations and seasonal adjustments in order to be modeled.

Based on the ARMA modeling, it seemed that ARMA(4,4) emerged as the best model with the lowest AIC (-5.060983) with the best statistical fit. The residuals also did not show any significant autocorrelation and residuals were approximately normal. This model needed manual de-trending and de-seasonalization since differencing could not be used, so a log-transformation was used and a seasonal means for de-seasonalization.

Comparing to the SARIMA modeling, it seems that SARIMA(2,0,0)x(0,1,0) was preferred for series with clear seasonal patterns, incorporating seasonal differencing (D=1) and periodicity (S=4). This model as well needed manual de-trending, which log-transformation was used, but differencing was used for de-seasonalizing. Eventhough this model had an AIC of (-4.306731), which is higher than the ARMA(4,4) model, the SARIMA model is better suited when seasonality plays a significant role.

Both ARMA and SARIMA models captured autocorrelation structures effectively but handled seasonality differently. The ARMA models only relied on AR and MA terms, while the SARIMA model incorporated seasonal differencing. As can be seen from the forecasts, the SARIMA(2,0,0)x(0,1,0) model is better for forecasting the quarterly revenue due to being able to handle seasonality, and with its robust structure it captured seasonality without overfitting.